

DAVID N. BELYN

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PROFESSIONAL SUMMARY

Analyst and strategist with 23+ years translating complex, ambiguous environments into actionable insight and quantitative decision support. Expertise spanning statistical modeling, risk assessment, strategic planning, and information operations across 15+ countries. Currently pursuing dual master's degrees in Applied Analytics and Applied Economics at Boston College (3.87 GPA), with graduate coursework in machine learning, econometrics, and statistical modeling. Recently designed and deployed an autonomous algorithmic trading and strategy-validation platform on live broker infrastructure, and published its out-of-sample findings. Combines quantitative rigor, hands-on Python, C++, and SQL engineering, and a record of disciplined judgment under uncertainty. Seeking to apply this background to quantitative modeling, analytics, and risk management within a financial institution.

CORE COMPETENCIES

Quantitative & Statistical Analysis • Econometric & Predictive Modeling • Backtesting & Out-of-Sample Validation • Algorithmic Trading Systems • Risk Assessment & Decision Support • Market & Operational Risk Concepts • Portfolio & Resource Management • Python, R, SQL, ML Pipelines • Data Visualization & Reporting • Executive Briefing & Presentations

EDUCATION

M.S. Applied Analytics & M.S. Applied Economics, Boston College | Expected Spring 2027

- Woods College of Advancing Studies. GPA 3.87. Coursework: machine learning, NLP, econometrics, statistical modeling, data visualization, game theory, industrial organization, market dynamics.

B.S. Business Administration, Finance, Auburn University | May 2011

- Magna Cum Laude with Honors.

QUANTITATIVE & RESEARCH PROJECTS

POPPER — Autonomous Trading & Strategy Validation Platform | 2026

- Designed and deployed a production algorithmic trading system against live broker infrastructure (OANDA v20): autonomous decision agent, deterministic server-side risk controls, streaming market data, and a real-time monitoring dashboard. Python, FastAPI, PostgreSQL, Docker.
- Built a strategy-validation pipeline using pre-registered acceptance criteria, out-of-sample splits, and Wilson confidence intervals; correctly rejected four candidate strategy families, including a data-mined signal that scored 77% in-sample and 15% out-of-sample.
- Implemented a lookahead-safe backtesting engine with transaction costs and financing accrual across 14+ years of daily data; verified engine integrity with a positive control that recovered the equity risk premium at Sharpe 0.55 net of costs.
- Engineered graduated execution tiers and hard pre-trade risk gates (exposure caps, volatility floors, event blackouts, daily-loss halt, kill switch) that contained two production defects, including a multi-container state-divergence failure, to under \$2 in realized cost.
- Published the methodology and negative results; codebase, incident analysis, and findings documented publicly.

MERIDIAN — Global Media Intelligence Dashboard | 2024–Present

- Built a full-stack Flask application using GDELT data to monitor global trends and narrative patterns across 100+ countries, deployed on Render.
- Applied BERTopic, transformer-based sentiment analysis, and custom frame scoring to surface emerging signals across sectors including fintech, defense technology, and logistics; methodology targeted for publication in *Parameters*.

Work-From-Home Equity & Labor Market Research | 2025

- Conducted primary analysis of IPUMS CPS microdata on remote work and employment equity; presented as a competitive poster at the BC Analytics & Industry Symposium. Findings challenged prevailing assumptions, demonstrating the ability to interrogate market narratives with data.

Capital One Case Competition — International Expansion Analysis | 2024

- Built market-entry recommendations for Vietnam and Brazil, synthesizing macroeconomic data, competitive dynamics, and risk considerations into executive-ready deliverables.

PROFESSIONAL EXPERIENCE

G39 Planner, Information Operations, 3rd Infantry Division | 2024–Present

- Lead strategic planning for a 20,000-person division, synthesizing intelligence on adversary messaging, information environment dynamics, and emerging technology risk.
- Oversee execution of a ~\$25M annual contracted maintenance program; track obligations and burn rates against approved funding and present funding recommendations to the Commanding General.

- Develop analytical decision-support products for general officer audiences, translating technical and operational complexity into clear frameworks.

Operations Advisor to Chief of General Staff, U.S. Military Training Mission, Saudi Arabia | 2023–2024

- Senior strategic advisor at the ministerial level for one of the largest U.S. defense partnerships, advising on modernization across a \$20B+ annual budget environment.
- Assessed risk and resource tradeoffs in a high-stakes, high-ambiguity environment, producing analysis consumed at the highest levels of government.

Battalion Executive Officer, 2nd Battalion, 47th Infantry Regiment | 2022–2023

- Second-in-command of a 600-person organization; managed budget execution, \$2.1M+ in equipment assets, and cross-functional staff operations.

Special Mission Unit Operational Planner, Special Operations Command Central | 2021–2022

- Deployed as forward planner; developed operational plans and provided real-time decision support to senior command under significant uncertainty.

Chief of Plans (S3), 198th Infantry Brigade | 2018–2020

- Directed brigade-level strategic planning; produced campaign plans aligned to Joint Chiefs and USSOUTHCOM direction and delivered analytical products for senior leadership.

Company Commander, 3-60th Infantry Regiment | 2015–2016

- Commanded a 240-soldier training company with full accountability for \$2.1M in equipment, sustaining a 98% completion rate across a 1,000+ person annual mission.

TECHNICAL SKILLS

Languages & Frameworks: Python (Pandas, NumPy, Scikit-learn, FastAPI, Flask, ML/NLP pipelines), R, SQL, C++, React/JavaScript

Modeling & Statistics: Econometric modeling, regression and classification (logistic regression, Linear SVM), time-series backtesting, hypothesis testing and confidence intervals, topic modeling (LDA/NMF, BERTopic), transformer models

Systems & Data: PostgreSQL, SQLAlchemy, Docker, REST and WebSocket APIs, Anthropic API (agentic tool use), OANDA v20, GDELT, IPUMS CPS

Analytics & Visualization: Tableau, ggplot2, matplotlib, Excel (advanced)

HONORS & RECOGNITION

Bronze Star Medal • Meritorious Service Medal • Joint Service Commendation Medal • Army Commendation Medal (5x) • Combat Infantryman's Badge • Expert Infantryman's Badge • Edward R. Murrow Award for Continuing Coverage (2002), Broadcast Journalism